

2(a)(i)	exocytosis ;	1															
2(a)(ii)	one mark per row	4															
	<table> <tr> <th>cell structure in Fig. 2.1</th><th>name of cell structure</th><th>function of cell structure in plasma cell</th></tr> <tr> <td>P</td><td>nucleolus</td><td>production / assembly, of (sub-units of) ribosomes ; A manufacture, ribosomal RNA / rRNA / transfer RNA / tRNA</td></tr> <tr> <td>Q</td><td>mitochondrion</td><td>R if answer refers to, ATP / energy, <u>for</u> aerobic respiration <u>aerobic respiration</u> ; A oxidative phosphorylation A provide, ATP / energy A produce / AW, <u>ATP</u></td></tr> <tr> <td>R</td><td>rough endoplasmic reticulum R rough ER / RER / ribosome</td><td>protein / antibody, synthesis / transport ; A synthesis of, polypeptide(s) / heavy and light chains A (post-translational) modification of, polypeptide / protein / antibody <i>allow example</i> A forms (transport) vesicles</td></tr> <tr> <td>S</td><td>Golgi body A Golgi, complex / apparatus</td><td>I lipids (post-translational) modification of, polypeptide / protein / antibody <i>allow example</i> or packaging, qualified ; e.g. packaging of antibodies / protein, into vesicles A formation of, vesicles / (primary) lysosomes</td></tr> </table>	cell structure in Fig. 2.1	name of cell structure	function of cell structure in plasma cell	P	nucleolus	production / assembly, of (sub-units of) ribosomes ; A manufacture, ribosomal RNA / rRNA / transfer RNA / tRNA	Q	mitochondrion	R if answer refers to, ATP / energy, <u>for</u> aerobic respiration <u>aerobic respiration</u> ; A oxidative phosphorylation A provide, ATP / energy A produce / AW, <u>ATP</u>	R	rough endoplasmic reticulum R rough ER / RER / ribosome	protein / antibody, synthesis / transport ; A synthesis of, polypeptide(s) / heavy and light chains A (post-translational) modification of, polypeptide / protein / antibody <i>allow example</i> A forms (transport) vesicles	S	Golgi body A Golgi, complex / apparatus	I lipids (post-translational) modification of, polypeptide / protein / antibody <i>allow example</i> or packaging, qualified ; e.g. packaging of antibodies / protein, into vesicles A formation of, vesicles / (primary) lysosomes	
cell structure in Fig. 2.1	name of cell structure	function of cell structure in plasma cell															
P	nucleolus	production / assembly, of (sub-units of) ribosomes ; A manufacture, ribosomal RNA / rRNA / transfer RNA / tRNA															
Q	mitochondrion	R if answer refers to, ATP / energy, <u>for</u> aerobic respiration <u>aerobic respiration</u> ; A oxidative phosphorylation A provide, ATP / energy A produce / AW, <u>ATP</u>															
R	rough endoplasmic reticulum R rough ER / RER / ribosome	protein / antibody, synthesis / transport ; A synthesis of, polypeptide(s) / heavy and light chains A (post-translational) modification of, polypeptide / protein / antibody <i>allow example</i> A forms (transport) vesicles															
S	Golgi body A Golgi, complex / apparatus	I lipids (post-translational) modification of, polypeptide / protein / antibody <i>allow example</i> or packaging, qualified ; e.g. packaging of antibodies / protein, into vesicles A formation of, vesicles / (primary) lysosomes															

Question	Answer	Marks
2(b)	any three from: 1 (named) small mammal, injected / AW, with antigen ; 2 immune response occurs (over several weeks) ; A immune response described 3 plasma cells / B-lymphocytes / B-cells / splenocytes, extracted from spleen ; 4 plasma cells / activated B-lymphocytes / activated B-cells, <u>fused</u> with, myeloma / tumour / cancer / AW, cells (to form hybridomas) ; 5 AVP ; e.g. hybridoma cells separated (into wells) to produce clones screening / selection / AW, for hybridomas producing desired, (monoclonal) antibodies / Mabs <i>ref. to large scale production</i>	3
2(c)	any four from: 1 (artificial) <u>passive immunity</u> ; R natural 2 no immune response ; I primary / secondary 3 <i>idea that</i> antibodies are, broken down / degraded / destroyed (in the body / blood) ; R 'antibodies die' 4 no long-term / only short-term / only temporary, immunity ; A no formation of immunological memory 5 memory cells not produced ; A memory cells only produced, in active immunity / during an immune response / after presentation of antigen 6 AVP ; e.g. <i>ref. to</i> no, clonal selection / clonal expansion / antibodies produced	4